

DEEP LEARNING

Ex 1 Code:

```
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import mnist
from sklearn.preprocessing import OneHotEncoder

(X_train, y_train), (X_test, y_test) = mnist.load_data()
X_train = X_train.reshape(-1, 784) / 255.0
encoder = OneHotEncoder(sparse_output=False)
y_train = encoder.fit_transform(y_train.reshape(-1, 1))

input_size, hidden_size, output_size = 784, 64, 10
W1 = np.random.randn(input_size, hidden_size) * 0.01
b1 = np.zeros((1, hidden_size))
W2 = np.random.randn(hidden_size, output_size) * 0.01
b2 = np.zeros((1, output_size))

sigmoid = lambda x: 1 / (1 + np.exp(-x))
sigmoid_deriv = lambda x: x * (1 - x)
loss_fn = lambda y, y_hat: -np.mean(y * np.log(y_hat + 1e-8))

epochs, lr = 10, 0.1
losses = []

for epoch in range(epochs):
    total_loss = 0
    for i in range(X_train.shape[0]):
        x = X_train[i:i+1]
        y = y_train[i:i+1]

        z1 = x @ W1 + b1
        a1 = sigmoid(z1)
        z2 = a1 @ W2 + b2
        a2 = sigmoid(z2)

        loss = loss_fn(y, a2)
        total_loss += loss

        dz2 = a2 - y
        dW2 = a1.T @ dz2
        db2 = dz2
        dz1 = (dz2 @ W2.T) * sigmoid_deriv(a1)
        dW1 = x.T @ dz1
        db1 = dz1

        W2 -= lr * dW2
        b2 -= lr * db2
        W1 -= lr * dW1
        b1 -= lr * db1

    losses.append(total_loss / X_train.shape[0])
    print(f"Epoch {epoch+1}, Loss: {losses[-1]:.4f}")

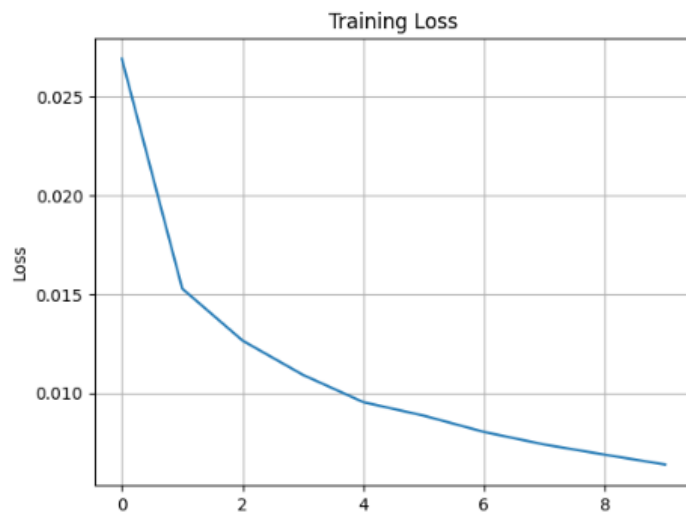
plt.plot(losses)
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
plt.show()

def predict(img):
    img = img.reshape(1, 784) / 255.0
    a1 = sigmoid(img @ W1 + b1)
    a2 = sigmoid(a1 @ W2 + b2)
    return np.argmax(a2)

idx = 100
plt.imshow(X_train[idx].reshape(28, 28), cmap='gray')
plt.title(f"Prediction: {predict(X_train[idx])}")
plt.axis('off')
plt.show()
```

OUTPUT:

```
*** Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz  
11490434/11490434 — 0s 0us/step  
Epoch 1, Loss: 0.0269  
Epoch 2, Loss: 0.0153  
Epoch 3, Loss: 0.0126  
Epoch 4, Loss: 0.0109  
Epoch 5, Loss: 0.0095  
Epoch 6, Loss: 0.0089  
Epoch 7, Loss: 0.0080  
Epoch 8, Loss: 0.0074  
Epoch 9, Loss: 0.0069  
Epoch 10, Loss: 0.0064
```



Prediction: 5

